

YIMING XIE

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EDUCATION

Northeastern University

Boston, MA, U.S.

- *Ph.D. Student, Khoury College of Computer Sciences*
- *Advisor: Huaizu Jiang*

Sep. 2021 - Jun. 2026 (expected)

Zhejiang University

Hangzhou, China

- *Bachelor of Opto-Electronics Information Science and Engineering*
- *Advisor: Xiaowei Zhou*

Sep. 2015 - June. 2019

WORK EXPERIENCE

3D Team, Stability AI

Remote

Research Scientist Intern

Feb 2024 - Dec 2024

Worked with Dr. Varun Jampani, Dr. Chun-Han Yao, and Dr. Vikram Voleti.

Worked on 4D generation.

Reality Labs Research, Meta

Redmond, WA, U.S.

Research Scientist Intern

May 2022 - Dec. 2022

Worked with Dr. Julian Straub, Prof. Georgia Gkioxari, and Dr. Minh Phuoc Vo.

Worked on multi-view 3D object detection.

State Key Laboratory of CAD&CG, Zhejiang University

Hangzhou, China

Research Assistant

Mar. 2021 - Aug. 2021

Worked with Prof. Xiaowei Zhou.

Worked on planar surface reconstruction.

MIG-3D&AR Group, SenseTime

Hangzhou, China

Research Intern

Nov. 2018 - Feb. 2021

Worked with Jiaming Sun and Prof. Xiaowei Zhou.

Worked on 3D scene reconstruction, 3D object detection and tracking, and scene flow estimation.

HONORS AND AWARDS

Apple Scholars in AI/ML

2024

CVPR Best Paper Candidate

2021

PREPRINTS

- Fangrui Zhu*, Hanhui Wang*, **Yiming Xie**, Jing Gu, Tianye Ding, Jianwei Yang, Huaizu Jiang. Struct2D: A Perception-Guided Framework for Spatial Reasoning in Large Multimodal Models. *In Submission*, 2025.
<https://arxiv.org/pdf/2506.04220>
- Zichong Meng, Zeyu Han, Xiaogang Peng, **Yiming Xie**, Huaizu Jiang. Absolute Coordinates Make Motion Generation Easy. *In Submission*, 2025.
<https://neu-vi.github.io/ACMDM/>

Note: * above denotes equal contribution

PUBLICATIONS

Google Scholar

Conferences

- Chun-Han Yao*, **Yiming Xie***, Vikram Voleti, Huaizu Jiang, Varun Jampani.
SV4D 2.0: Enhancing Spatio-Temporal Consistency in Multi-View Video Diffusion for High-Quality 4D Generation.
International Conference on Computer Vision (ICCV), 2025.
<https://sv4d20.github.io/>
- Zichong Meng, **Yiming Xie**, Xiaogang Peng, Zeyu Han, Huaizu Jiang.
Rethinking Diffusion for Text-Driven Human Motion Generation.
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2025.
<https://neu-vi.github.io/MARDM/>
- Xiaogang Peng*, **Yiming Xie***, Zizhao Wu, Varun Jampani, Deqing Sun, Huaizu Jiang.
HOI-Diff: Text-Driven Synthesis of 3D Human-Object Interactions using Diffusion Models.
HuMoGen Workshop@CVPR, 2025.
<https://neu-vi.github.io/HOI-Diff/>
- **Yiming Xie***, Chun-Han Yao*, Vikram Voleti, Huaizu Jiang, Varun Jampani.
SV4D: Dynamic 3D Content Generation with Multi-Frame and Multi-View Consistency.
International Conference on Learning Representations (ICLR), 2025.
<https://sv4d.github.io/>
- Lei Zhong, **Yiming Xie**, Varun Jampani, Deqing Sun, Huaizu Jiang.
SMooDi: Stylized Motion Diffusion Model.
The European Conference on Computer Vision (ECCV), 2024.
<https://neu-vi.github.io/SMooDi/>
- **Yiming Xie**, Varun Jampani, Lei Zhong, Deqing Sun, Huaizu Jiang.
OmniControl: Control Any Joint at Any Time for Human Motion Generation.
International Conference on Learning Representations (ICLR), 2024.
<https://neu-vi.github.io/omnicontrol/>
- **Yiming Xie**, Huaizu Jiang, Georgia Gkioxari, Julian Straub.
Pixel-Aligned Recurrent Queries for Multi-View 3D Object Detection.
International Conference on Computer Vision (ICCV), 2023.
<https://ymingxie.github.io/parq/>
- **Yiming Xie**, Matheus Gadelha, Fengting Yang, Xiaowei Zhou, Huaizu Jiang.
PlanarRecon: Real-time 3D Plane Detection and Reconstruction from Posed Monocular Videos.
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022.
<https://neu-vi.github.io/planarrecon/>
- **Yiming Xie***, Jiaming Sun*, Siyu Zhang, Linghao Chen, Xiaowei Zhou.
You Don't Only Look Once: Constructing Visuospatial Working Memory for Integrated 3D Object Detection and Tracking.
International Conference on Computer Vision (ICCV), 2021.
(**Strong Accept * 2 in final rating**)
<https://zju3dv.github.io/udolo/>
- **Yiming Xie***, Jiaming Sun*, Linghao Chen, Xiaowei Zhou.
NeuralRecon: Real-Time Coherent 3D Reconstruction from Monocular Video.
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2021.
(**Oral and Best Paper Candidate**)
<https://zju3dv.github.io/neuralrecon/>
- Jiaming Sun*, Linghao Chen*, **Yiming Xie**, Siyu Zhang, Qinhong Jiang, Xiaowei Zhou, Hujun Bao.

Disp R-CNN: Stereo 3D Object Detection via Shape Prior Guided Instance Disparity Estimation.
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2020.
<https://arxiv.org/abs/2004.03572>

Note: * above denotes equal contribution

Journals

- Lei Zhong, Chuan Guo, **Yiming Xie**, Jiawei Wang, Changjian Li.
Sketch2Anim: Towards Transferring Sketch Storyboards into 3D Animation.
ACM Transactions on Graphics (TOG) (Proc. SIGGRAPH), 2025.
<https://zhongleilz.github.io/Sketch2Anim/>
- Xi Chen, Jiaming Sun, **Yiming Xie**, Hujun Bao, Xiaowei Zhou.
NeuralRecon: Real-Time Coherent 3D Scene Reconstruction from Monocular Video.
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2024.
<https://ieeexplore.ieee.org/document/10508111>
- Linghao Chen*, Jiaming Sun*, **Yiming Xie**, Siyu Zhang, Qing Shuai, Qinhong Jiang, Guofeng Zhang, Hujun Bao, Xiaowei Zhou.
Shape Prior Guided Instance Disparity Estimation for 3D Object Detection.
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2021.
<https://ieeexplore.ieee.org/document/9419782>
- Fangrui Zhu, **Yiming Xie**, Weidi Xie, Huaizu Jiang.
Diagnosing Human-object Interaction Detectors.
International Journal of Computer Vision (IJCV), 2025.
<https://neu-vi.github.io/Diag-HOI/>

PROFESSIONAL SERVICE

- Conference Reviewer: CVPR, ICCV, ECCV, NeurIPS, ICLR, ICML, SIGGRAPH
- Journal Reviewer: TVCG, TCSVT, EAAI

TEACHING EXPERIENCE

- Teaching assistant of CS 7150, Deep Learning, Northeastern University Spring 2022/2023

TALKS

Dynamic 3D Content Generation. GAMES	Aug. 2024
Dynamic 3D Content Generation with Multi-Frame and Multi-View Consistency. BAAI	Aug. 2024
NeuralRecon: Real-Time 3D Reconstruction from Monocular Video. CSIG-3DV	Aug. 2021
Real-Time 3D Reconstruction from Monocular Video. AI Time	Apr. 2021

SKILLS

Programming Languages: C, C++, Python, LaTeX

Technical: Pytorch, Git, Linux, Blender, Final cut pro